

Data Quality Report

AI Workforce Intelligence 2030
Phase 2 – Data Audit & Profiling

1. Objective

The objective of this data quality assessment is to evaluate the completeness, consistency, validity, and overall reliability of the dataset before proceeding with Exploratory Data Analysis (EDA). Ensuring high data quality at this stage minimizes analytical errors and increases confidence in the business insights generated during subsequent phases.

2.Dataset Overview

Attribute	Details
Dataset Name	AI Impact on Jobs Market in 2030
Total Records	3000
Total Values	20
File Format	.csv
Data Type	Structured Tabular Dataset

3. Data Quality Assessment

Quality Check	Result	Observation
Missing Values	Passed	No missing values were identified across any of the dataset columns.
Duplicate Values	Passed	No duplicate records were detected. Each employee's record is unique.
Data Types	Passed	Numerical and categorical variables were stored using appropriate data types for analysis.
Unique Values	Passed	Categorical variables contained valid and meaningful unique values suitable for business analysis.
Dataset Structure	Passed	The dataset consists of 20 well-defined variables covering workforce, AI adoption, compensation, skills, and employment trends.

4. Data Quality Summary

The data profiling exercise indicates that the dataset is complete, internally consistent, and suitable for analytical purposes. No significant data quality issues were identified that would negatively affect exploratory analysis or dashboard development.

Since the dataset contains no missing values or duplicate observations and follows an appropriate structural format, extensive preprocessing is not required before analysis.

5. Limitations

Although the dataset demonstrates good technical quality, the following analytical limitations should be considered:

- The dataset is cross-sectional and represents a single snapshot rather than historical trends.
- Causal relationships cannot be established solely from this dataset.
- Forecasting future workforce outcomes would require historical or time-series data.

6. Conclusion

Based on the data audit and profiling activities, the dataset has successfully passed the preliminary quality assessment. The information is complete, structurally consistent, and analysis-ready. Only optional feature engineering may be performed during later stages if required to improve interpretability or support specific business questions.